

H1 SF2525

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1 Proof 1

We want to prove that:

$$\int_0^T W_t dW_t = \frac{W_T^2}{2} - \frac{T}{2}$$

Knowing that the limit of the discretization made with the forward Euler method defines the Ito integral, we can write:

$$\int_0^T W_t dW_t = \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} W(t_n) \Delta W_n$$

where $0 = t_1 < t_2 < \dots < t_N = T$ is a partition of the interval $[0, T]$ and we define the time increments $\Delta t_n = (t_{n+1} - t_n)$ and $\Delta W_n = (W(t_{n+1}) - W(t_n))$ for $n = 1, 2, \dots, N-1$.

In order to prove the original equality, we want to show that, when $N \rightarrow \infty$:

$$E[(\sum_{n=0}^{N-1} W(t_n) \Delta W(t_n) - (\frac{W_T^2}{2} - \frac{T}{2}))^2] \rightarrow 0$$

Notice that we are verifying the L^2 convergence in this way (in fact we know that the forward Euler method discretizations converge in L^2).

We start by noticing that we can rewrite the summation in the following way:

$$\sum_{n=0}^{N-1} W(t_n) \Delta W(t_n) = \sum_{n=0}^{N-1} W(t_n) (W(t_{n+1}) - W(t_n)) = \sum_{n=0}^{N-1} \frac{W^2(t_{n+1}) - W^2(t_n)}{2} - \frac{(\Delta W_n^2)}{2}$$

Substituting these terms in the expected value:

$$\begin{aligned} E[(\sum_{n=0}^{N-1} \frac{W^2(t_{n+1}) - W^2(t_n)}{2} - \frac{(\Delta W_n^2)}{2} - (\frac{W_T^2}{2} - \frac{T}{2}))^2] = \\ E[(\frac{W_T^2 - W^2(0)}{2} - \sum_{n=0}^{N-1} \frac{(\Delta W_n^2)}{2} - (\frac{W_T^2}{2} - \frac{T}{2}))^2] \end{aligned}$$

Since by definition of Wiener process, $W(0) = 0$, then:

$$E[(\frac{W_T^2}{2} - \sum_{n=0}^{N-1} \frac{(\Delta W_n^2)}{2} - (\frac{W_T^2}{2} - \frac{T}{2}))^2] = E[(\sum_{n=0}^{N-1} \frac{(\Delta W_n^2)}{2} - \frac{T}{2})^2] =$$

In order to write the argument of the expected value as a summation, we can write $T = \sum_{n=0}^{N-1} \Delta t_n$ and knowing that $\Delta W_n \sim \mathcal{N}(0, \Delta t_n) \forall n = 1, \dots, N-1$ and that they are independent and using linearity property of the expected value, it holds:

$$\begin{aligned} E[(\sum_{n=0}^{N-1} \frac{(\Delta W_n^2)}{2} - \frac{T}{2})^2] &= \frac{1}{4} \sum_{n=0}^{N-1} E[(\Delta W_n^2 - \Delta t_n)^2] = \\ &= \frac{1}{4} \sum_{n=0}^{N-1} E[\Delta W_n^4 + \Delta t_n^2 - 2\Delta W_n^2 \Delta t_n] = \frac{1}{4} \sum_{n=0}^{N-1} E[\Delta W_n^4] + \Delta t_n^2 - 2\Delta t_n E[\Delta W_n^2] \end{aligned}$$

Here we compute $E[\Delta W_n^4] = 3\Delta t_n^2$ and $E[\Delta W_n^2] = \Delta t_n$ and plug in the results, obtaining:

$$\begin{aligned} \frac{1}{4} \sum_{n=0}^{N-1} 3\Delta t_n^2 + \Delta t_n^2 - 2\Delta t_n^2 &= \\ &= \frac{1}{2} \sum_{n=0}^{N-1} \Delta t_n^2 \end{aligned}$$

Finally, we got a quantity whose limit (for $N \rightarrow \infty$) goes to 0 (notice that $\Delta t_n \leq \Delta t_{max} \forall n$, where the maximum time step $\Delta t_{max} = c \frac{T}{N}$ and $c \in R$ is a constant, so:

$$\Delta t_n \leq \Delta t_{max} = c \frac{T}{N} \rightarrow 0$$

Therefore, we proved that

$$\int_0^T W_t dW_t = \frac{W_T^2}{2} - \frac{T}{2}$$

2 Proof 2

We want to prove that:

$$E[(\int_0^T f(t, \cdot) dW_t)(\int_0^T g(t, \cdot) dW_t)] = \int_0^T E[f(t, \cdot)g(t, \cdot)] dt$$

Knowing that the limit of the discretization made with the forward Euler method defines the Ito integral, we compute the discretizations of both the integrals in the LHS and then compute its limit.

Discretizing the two integrals in the left handside of the equation, we get:

$$E[\sum_{n=0}^{N-1} f(t_n, \cdot) \Delta W_n \sum_{n=0}^{N-1} g(t_n, \cdot) \Delta W_n]$$

Notice that we can consider the same time partition for both integrals without loss of generality because given two different forward Euler approximations I_1 and I_2 , it holds:

$$\lim_{\Delta t_{max} \rightarrow 0} E[(I_1 - I_2)^2] \rightarrow 0$$

where Δt_{max} is the maximum time step between the two partitions (and in the conclusion of the proof we will have to compute the limit for $\Delta t_n \rightarrow 0$ (which obviously implies $\Delta t_{max} \rightarrow 0$) to pass from the discretization to the continuous case.

We can rewrite the discretization as follows:

$$E\left[\sum_{n=0}^{N-1} f(t_n, \cdot)g(t_n, \cdot)\Delta W_n^2 + \sum_{n \neq m} f(t_n, \cdot)g(t_m, \cdot)\Delta W_n\Delta W_m\right]$$

then, thanks to linearity of the expected value and independence of increments ΔW :

$$\sum_{n=0}^{N-1} E[f(t_n, \cdot)g(t_n, \cdot)]E[\Delta W_n^2] + \sum_{n \neq m} E[f(t_n, \cdot)g(t_m, \cdot)]E[\Delta W_n]E[\Delta W_m]$$

Since $\Delta W \sim \mathcal{N}(0, \Delta t_n)$, the previous terms are equal to:

$$\sum_{n=0}^{N-1} E[f(t_n, \cdot)g(t_n, \cdot)]\Delta t_n$$

And taking the limit of this discretization for $N \rightarrow \infty$ (and consequently $\Delta t_n \rightarrow 0$) we get:

$$\sum_{n=0}^{N-1} E[f(t_n, \cdot)g(t_n, \cdot)]\Delta t_n \rightarrow \int_0^T E[f(t, \cdot)g(t, \cdot)] dt$$

Actually, the previous relationship only holds if

$$E[f(t, \cdot)g(t, \cdot)]$$

is a continuous function. Let's analyse the following expression to show continuity:

$$\begin{aligned} & |E[f(t + \Delta t, \cdot)g(t + \Delta t, \cdot) - f(t, \cdot)g(t, \cdot)]| = \\ & = |E[f(t + \Delta t, \cdot)g(t + \Delta t, \cdot) - f(t, \cdot)g(t + \Delta t, \cdot) + f(t, \cdot)g(t + \Delta t, \cdot) - f(t, \cdot)g(t, \cdot)]| \leq \\ & \leq |E[f(t + \Delta t, \cdot)g(t + \Delta t, \cdot) - f(t, \cdot)g(t + \Delta t, \cdot)]| + |E[f(t, \cdot)g(t + \Delta t, \cdot) - f(t, \cdot)g(t, \cdot)]| \leq \\ & \leq \sqrt{E[(f(t + \Delta t, \cdot)g(t + \Delta t, \cdot) - f(t, \cdot)g(t + \Delta t, \cdot))^2]} + \sqrt{E[(f(t, \cdot)g(t + \Delta t, \cdot) - f(t, \cdot)g(t, \cdot))^2]} = \\ & = \sqrt{E[g(t + \Delta t, \cdot)^2]E[(f(t + \Delta t, \cdot) - f(t, \cdot))^2]} + \sqrt{E[f(t, \cdot)^2]E[(g(t + \Delta t, \cdot) - g(t, \cdot))^2]} \leq \\ & \leq C\sqrt{\Delta t}(\sqrt{E[g(t + \Delta t, \cdot)^2]} + \sqrt{E[f(t, \cdot)^2]}) \end{aligned}$$

where first and second inequalities are respectively triangle inequality and Cauchy-Schwartz inequality, while the third inequality comes from the initial hypothesis of the theorem (C is a constant). Now, if the quantity

$$\sqrt{E[g(t + \Delta t, \cdot)^2]} + \sqrt{E[f(t, \cdot)^2]}$$

can be bounded in the time interval $[0, T]$ by a certain constant, then we have proven the continuity of

$$E[f(t, \cdot)g(t, \cdot)]$$